Error Handling & Data Validation

Module 1 — Understanding Excel Errors

Why Error Handling Matters

In business, Excel reports often contain thousands of rows imported from ERP, CRM, SQL databases, or CSV files. Missing values, incorrect formulas, deleted references, and invalid lookups can cause errors.
A Data Analyst must understand:

- Why an error occurs
- How to identify it
- How to fix it
- How to prevent it

Proper error handling ensures accurate reporting, clean dashboards, and reliable business insights.

Common Excel Errors

Error	Meaning	Common Reason
#N/A	Value not found	Lookup value missing
#VALUE!	Wrong data type	Text used instead of number
#REF!	Invalid cell reference	Deleted rows/columns
#DIV/0!	Division by zero	Blank or zero denominator
#NAME?	Unknown function/name	Typing mistake
#NUM!	Invalid numeric value	Impossible numeric calculation
#NULL!	Incorrect range intersection	Wrong range operator
#####	Column too narrow	Width insufficient or negative date

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1. #N/A Error

Occurs when Excel cannot find the required lookup value.

Business Example

Product ID does not exist in Product Master.
P-2050
Master table contains only
P-1001
P-1002
P-1003
VLOOKUP returns
#N/A

Formula
=VLOOKUP(A2,$H$2:$J$20,2,FALSE)

Business Use Case

- Product lookup
- Customer lookup
- Employee lookup
- Vendor lookup

Business Solution

=IFNA(VLOOKUP(A2,$H$2:$J$20,2,FALSE),"Product Not Found")

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2. #VALUE! Error

Occurs when Excel receives an incorrect data type.
Example:
Trying to add text with numbers.

Incorrect
="Laptop"+200
Returns
#VALUE!

Correct
=VALUE(A2)+200

Business Use Cases

- Imported CSV
- ERP exports
- Sales data
- Invoice amounts stored as text

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3. #REF! Error

Occurs when a referenced cell no longer exists.

Example
Original Formula
=B2+C2
Delete Column C
Result
#REF!

Business Scenario

Deleting source columns after creating reports.

Prevention
Avoid deleting referenced columns without checking dependent formulas.

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4. #DIV/0!

Occurs when dividing by zero.

Example
=Sales/Quantity
Quantity
0
Result
#DIV/0!

Business Example

- Average Revenue
- Profit Margin
- Cost per Unit

Solution
=IF(B2=0,"N/A",A2/B2)
or
=IFERROR(A2/B2,0)

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5. #NAME?

Excel cannot recognize the function.

Example
=SUMM(A1:A10)
Returns
#NAME?

Correct
=SUM(A1:A10)

Business Cause

- Typing mistakes
- Incorrect named ranges

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6. #NUM!

Occurs when Excel receives an invalid numeric value.

Example
=SQRT(-25)
Returns
#NUM!

Business Example

- Financial forecasting
- Interest calculations
- Loan calculations

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7. #NULL!

Occurs when the intersection operator is used incorrectly.

Example
=A1:A5 C1:C5
Returns
#NULL!

Usually happens due to typing mistakes.

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8. ##### Error

Not actually a formula error.
Usually means

- Column width too small
- Negative date/time

Solution
Increase column width.

Module 2 — Error Handling Functions

IFERROR Function

Returns a custom value instead of an Excel error.

Syntax
=IFERROR(value,value_if_error)

Arguments
value
Formula to evaluate.
value_if_error

Value returned when an error occurs.

Example
=IFERROR(A2/B2,0)

Business Example
Sales Report
Instead of
#DIV/0!
Display
0

Business Uses

- Dashboards
- KPI Reports
- Financial Reports
- Sales Reports

IFNA Function

Handles only
#N/A

Syntax
=IFNA(value,value_if_na)

Example
=IFNA(XLOOKUP(A2,H:H,I:I),"Not Available")

Business Example

- Product not found
- Customer not found
- Employee not found

Difference

IFERROR	IFNA
Handles all errors	Handles only #N/A

ERROR.TYPE Function

Returns a numeric code representing the error.

Syntax
=ERROR.TYPE(A2)

Returns

Error	Code
#NULL!	1
#DIV/0!	2
#VALUE!	3
#REF!	4
#NAME?	5
#NUM!	6
#N/A	7

Business Use

- Audit reports
- Formula debugging
- Large workbook troubleshooting

ISERROR Function

Checks whether a value contains any Excel error.

Syntax
=ISERROR(A2)

Returns
TRUE
or
FALSE

Business Example

- Identify broken formulas before sharing reports.

ISNA Function

Checks specifically for
#N/A

Syntax
=ISNA(A2)

Business Example

- Missing Product IDs
- Missing Customers

ISERR Function

Checks every error except
#N/A
Useful when lookup failures should remain visible but all other errors should be flagged.

Syntax
=ISERR(A2)

Module 3 — Data Validation

What is Data Validation?

Data Validation restricts users to entering only valid and expected data, reducing errors at the point of data entry.

Why Data Analysts Use Data Validation

- Prevent incorrect data entry
- Standardize datasets
- Reduce cleaning effort
- Improve reporting accuracy
- Ensure business rules are followed

Types of Data Validation

1. Whole Number

Allow only integers.
Example:
Age
18–60

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2. Decimal

Allow decimal values.
Example
Discount %
0–50%

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3. Date

Allow only valid dates.
Example
Joining Date
01-Jan-2025
to

31-Dec-2025

4. Time

Allow office timings only.
09:00 AM
to
06:00 PM

5. List

Create dropdown menus.
Example
Department
Finance

HR

Sales

IT

Marketing

6. Text Length

Restrict character count.
Employee ID
Exactly
8 characters

7. Custom Formula

Create business rules.
Example
Quantity
Must be greater than zero.
Formula
=A2>0

Input Message

Displays instructions before data entry.
Example
Enter Quantity between 1 and 100

Error Alert

Custom message shown when invalid data is entered.
Example
Invalid Product ID

Business Benefit
Reduces data-entry mistakes before they reach reports.

Real Business Use Cases

Sales
Only approved product categories.

HR
Joining Date cannot be in the future.

Finance
Discount cannot exceed 30%.

Inventory
Quantity cannot be negative.

Banking
Loan amount must be positive.

Healthcare
Patient Age
0–120 years only.

Interview Questions

Q1. What is the difference between IFERROR and IFNA?
Answer: IFERROR handles all Excel errors (#N/A, #VALUE!, #REF!, #DIV/0!, etc.), whereas IFNA handles only the #N/A error. Use IFNA specifically for lookup functions when a missing match is expected.

Q2. Which error is most common with VLOOKUP and XLOOKUP?
Answer: #N/A, because the lookup value is not found in the lookup table.

Q3. How do you avoid #DIV/0! errors?
Answer: Check if the denominator is zero using IF, or use IFERROR to return a custom value instead of the error.

Q4. What causes a #REF! error?
Answer: A formula refers to a cell, row, or column that has been deleted or no longer exists.

Q5. Why is Data Validation important in business?
Answer: It prevents invalid data entry, improves data quality, reduces cleaning effort, and ensures reports are accurate and consistent.

Q6. What is the purpose of ERROR.TYPE()?
Answer: It returns a numeric code identifying the type of Excel error, making it useful for auditing and troubleshooting large workbooks.

Q7. When would you use a List validation?
Answer: To restrict users to predefined values, such as Departments, Regions, Payment Methods, or Product Categories.

Q8. Explain the difference between ISERROR() and ISNA().
Answer: ISERROR() checks for any Excel error, while ISNA() checks only for the #N/A error.

Q9. Give one business example where IFERROR improves reporting.
Answer: In a sales dashboard, replacing #DIV/0! or #N/A with 0 or "Data Not Available" keeps KPI reports clean and easier for stakeholders to understand.

Q10. Why should analysts prefer Data Validation over fixing errors later?
Answer: Preventing incorrect input is more efficient than cleaning bad data afterward. It saves time, improves accuracy, and reduces the risk of incorrect business decisions.
This lecture structure aligns well with a job-oriented Excel curriculum and covers the error handling and validation concepts commonly expected in Data Analyst interviews before progressing to Pivot Tables, Power Query, or dashboard development.
